

AMIR MOUSAVI

AI / MACHINE LEARNING ENGINEER | COMPUTER VISION | RESEARCHER

France • Email | GitHub | LinkedIn | Website | Google Scholar

PROFILE

AI / Machine Learning Engineer, Data Scientist, and Researcher with a strong mathematics and engineering background, driven by curiosity and a genuine interest in solving complex, open-ended problems. I enjoy learning new concepts, exploring unfamiliar ideas, and working through challenging problems that require deep thinking, experimentation, and persistence. I am especially motivated by problems where there is no obvious solution; understanding them from first principles, evaluating different approaches, and iterating until a practical and reliable solution emerges. My experience spans machine learning, deep learning, computer vision, statistical modeling, software engineering, and neuroscience, from data analysis, model development, and GPU-accelerated training to RAG systems, APIs, data pipelines, and deployment. I have also worked in multidisciplinary and international collaborations, contributing across research, engineering, clinical, and software-focused teams. I combine research-oriented thinking, engineering execution, and collaborative problem solving, with a strong willingness to learn, exchange ideas, and turn complex concepts into reproducible, scalable, and useful AI systems.

CORE SKILLS

Applied ML and Data Science

Python, scikit-learn, XGBoost, SVMs, regression, ensemble learning, statistical/Bayesian analysis, graph-based ML, cross-validation, optimization

Deep Learning and Vision

TensorFlow, CNNs, GNNs, image segmentation, object detection, classification, GANs, medical imaging, 3D/volumetric imaging

Generative and Agentic AI

RAG, LLM applications, embeddings, semantic chunking, vector/keyword/hybrid retrieval, grounded QA, agentic frameworks, deep reinforcement learning

Software and Infrastructure

FastAPI, Django, Flask, Node.js, Express, Next.js, React, ROS, REST APIs, PostgreSQL, Redis, OpenSearch, MongoDB, Docker, Nginx, Linux, Git, GitHub Actions, CI/CD, CUDA, cuDNN

PROFESSIONAL EXPERIENCE

Machine Learning Researcher / PhD Candidate | Université de Picardie Jules Verne (UPJV)

Amiens, France | 2023 - Expected Sep 2026

- Develop and evaluate SVM, regression, ensemble, graph-based, and voxel-decoding machine-learning models using brain-connectivity and clinical data from approximately **400 stroke patients** to investigate and predict cognitive and motor outcomes.
- Build reproducible Python ML pipelines for preprocessing, feature engineering, hyperparameter optimization, cross-validation, statistical/Bayesian analysis, and model interpretation.
- Collaborate with multidisciplinary research and clinical teams; presented work at NetSci 2025 (Maastricht) and the OHBM 2026 Annual Meeting (Bordeaux), with a related manuscript under review.

Lead Developer & Architect | DocuMind & PaperMind

Scientific RAG platform | 2026 | Email | Email

- Architected and developed a self-hosted scientific RAG platform for organizing PDFs, retrieving evidence, and answering grounded questions with source citations.
- Implemented document extraction, semantic chunking, embeddings, and vector, keyword, and hybrid retrieval using OpenSearch and local model inference.
- Designed provenance-aware asynchronous pipelines integrating Python services, Node.js, Express, Next.js, PostgreSQL, Redis, and REST APIs; implemented authentication, Docker/Nginx deployment, and CI/CD.

Teaching Assistant - Python & Machine Learning | Université de Picardie Jules Verne (UPJV)

UPJV, Amiens, France | 2024 - 2026

- Delivered approximately 60 hours of university-level Python and machine-learning teaching, laboratory support, debugging assistance, and student guidance.

Backend Developer | Maralhost

Isfahan, Iran | 2021 - 2022

- Developed and maintained Django-based backend services and API microservices for web and server-management applications, including relational data models and server-side business logic.
- Designed authentication and authorization mechanisms including Single Sign-On (SSO), and supported Linux-based deployment, debugging, database-backed services, web vulnerability assessment, and collaborative Git workflows.

Programmer & Electronics Designer - Embedded Systems | Mobarakeh Steel Company

Isfahan, Iran | 2017 - 2019

- Analyzed and redesigned an electronic control board for rolling-equipment speed regulation, integrating a microcontroller-based control system.
- Developed control algorithms and evaluated system performance, behavior, and reliability under operational conditions.

PROJECTS

Agentic AI for Adaptive 3D Stroke Lesion Segmentation

Agentic AI | Deep RL | Computer Vision | 2026 - Present

- Developing an agentic AI framework for adaptive 3D stroke-lesion segmentation integrating segmentation models, uncertainty estimation, lesion-level analysis, and sequential specialist-model routing.
- Designing comparisons between rule-based routing, learned policy networks, multimodal LLM agents, and reinforcement-learning controllers, with planned policy optimization using segmentation quality, uncertainty, and computational cost as reward signals.

GPU-Accelerated Brain Ischemic Stroke Segmentation

Deep Learning | Medical Imaging | GPU Computing | 2025 - 2026

- Developed novel deep-learning segmentation models (RAX-NET and VRXU-net preprints) for ischemic stroke lesions in T1-weighted brain MRI using attention and residual mechanisms.
- Built GPU-accelerated TensorFlow training pipelines using CUDA/cuDNN for preprocessing, batching, training, and experimentation on volumetric medical-imaging data.

Neural Dynamics of Conventional and Virtual Reality Trail Making: A Source-Imaging EEG Study

VR | EEG | Research Software Collaboration | 2025

- Collaborated on virtual-reality implementations of the Trail Making Test A/B (TMT-A/TMT-B) using a Meta VR headset for cognitive-assessment experiments.
- Developed and integrated experimental tools for VR task presentation and EEG acquisition, connecting VR interaction, cognitive-task execution, and neurophysiological data collection.

Lumbar Spine Degeneration Classification

Deep Learning | Medical Imaging | 2024

- Developed a deep-learning approach for automated lumbar-spine degeneration classification using YOLOv8 and DeepScoreNet.
- Built and evaluated a computer-vision pipeline for identifying relevant spinal structures and classifying degenerative conditions.

Synthetic Face Generation Through Eyes-to-Face Inpainting

Computer Vision | Generative Modeling | Biometrics | IEEE IJCB 2023

- Developed an eyes-to-face inpainting approach that generates synthetic facial images by combining eye regions from different individuals and reconstructing complete faces.
- Evaluated synthetic datasets for age/gender estimation and face-recognition training across four benchmark datasets, and investigated demographic bias and underrepresented groups.

Deep Reinforcement Learning in Secure Wireless Communication

Research Collaboration | WPI | 2022 - 2023

- In a research collaboration with Worcester Polytechnic Institute (WPI), implemented a Deep Deterministic Policy Gradient (DDPG) algorithm for secure wireless communication.
- Applied the DDPG-based reinforcement-learning approach to improve wireless transmission performance.

Baxter Chess-Playing Robot - Vision-Based Robotic Chess System

Computer Vision | Robotics | Software Engineering | 2026

- Developed a robotic chess system integrating Kinect-based computer vision, chess-state reasoning, Stockfish, FastAPI, ROS, calibration, and motion control.
- Implemented the workflow for detecting human moves, updating board state, generating the robot's response, and commanding a Baxter robot to manipulate chess pieces.

COVID-19 Lung & Lesion Segmentation

CT Imaging | Segmentation | Web Deployment | 2020 - 2021

- Developed a deep-learning pipeline for automated lung and COVID-19 lesion segmentation from chest CT scans using clinical imaging data collected from hospitals.
- Trained segmentation models to quantify affected lung regions and integrated preprocessing, inference, and visualization into a web-based application for healthcare use.

Real-Time Face-Mask Detection on Raspberry Pi

Edge AI | Computer Vision | 2020

- Developed a real-time deep-learning system for detecting face-mask usage using a Raspberry Pi camera.
- Implemented image acquisition, preprocessing, inference, and visualization for resource-constrained edge hardware.

Persian NLP for E-commerce Review Analysis

Bachelor Thesis | NLP | Machine Learning | 2019

- Developed a Persian NLP pipeline using approximately 15,000 Digikala customer reviews.
- Built TF-IDF + Linear SVM text classification in scikit-learn and implemented sentiment-analysis and LDA topic-modeling workflows for positive/negative product feedback.

EDUCATION AND ADDITIONAL INFORMATION

PhD Candidate in Neuroscience | Université de Picardie Jules Verne (UPJV) Amiens, France | 2023 - 2026

Research focus: machine learning and data-science methods for stroke and functional connectivity, statistical modeling, graph analysis, optimization, cross-validation, and reproducible scientific computing.

MSc in Electrical Engineering - Telecommunications | Shiraz University of Technology Shiraz, Iran | 2019 - 2023

Thesis: Quantification of damaged lung tissue due to COVID-19 using computed tomography images.

BEng in Electrical Engineering - Telecommunications | University of Isfahan Isfahan, Iran | 2015 - 2019

Final project: Persian NLP for e-commerce review classification, sentiment analysis, and topic modeling.

SELECTED PUBLICATION

Mousavi Mobarakeh, S. A. VRXU-net: A Deep Learning Approach for Brain Ischemic Stroke Lesion Detection and Segmentation in T1W MRI. arXiv, 2026. [DOI](#)

Mousavi, A. RAX-NET: Residual Attention Xception Network for Brain Ischemic Stroke Segmentation in T1-Weighted MRI. medRxiv, 2025. [DOI](#)

Mousavi, A. Lumbar Spine Degenerative Classification Using YOLO v8 and DeepScoreNet. medRxiv, 2024. [DOI](#)

Mousavi Mobarakeh, S. A.; Kazemi, K.; Aarabi, A.; Danyal, H. Empowering Medical Imaging with Artificial Intelligence: A Review of Machine Learning Approaches for Detection and Segmentation of COVID-19 Using Radiographic and Tomographic Images. arXiv, 2024. [DOI](#)

Hassanpour, A.; Mousavi Mobarakeh, S. A.; Etefaghi Daryani, A.; Ramachandra, R.; Yang, B. Synthetic Face Generation Through Eyes-to-Face Inpainting. IEEE IJCB, 2023. [DOI](#)

Full publication list: [Google Scholar](#)

ADDITIONAL INFORMATION

Conferences: NetSci 2025 (Maastricht); OHBM 2026 Annual Meeting (Bordeaux).

Languages: English (professional proficiency); French (B1 listening, A2 speaking, A1 writing).